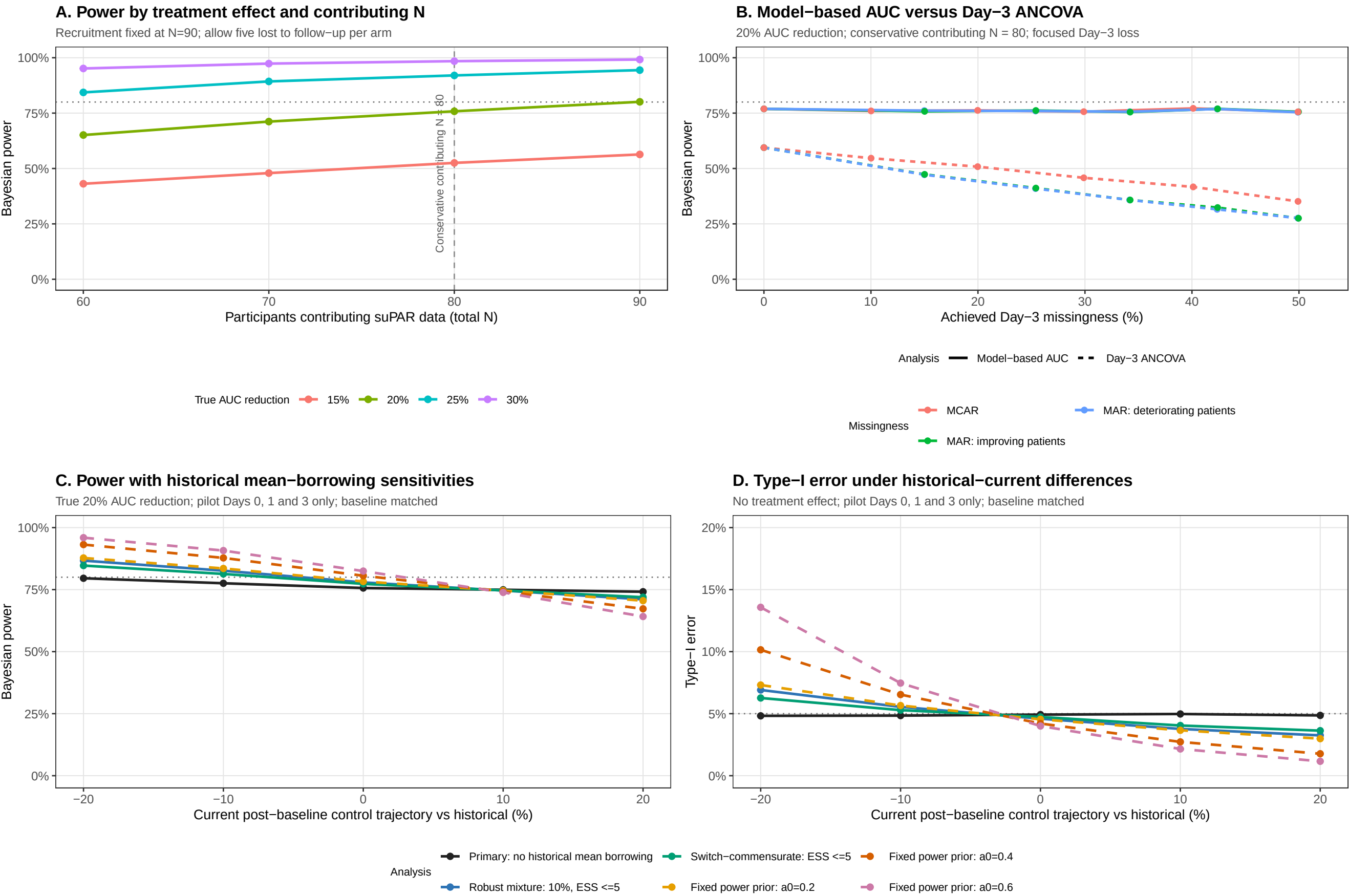


Dexamethasone–suPAR trial: operating characteristics supporting the primary analysis

Primary decision: $P(\Delta < 0 \mid \text{data}) \geq 0.95$, $\Delta = \log(\text{AUC dexamethasone}/\text{AUC control})$; common post-baseline treatment-effect prior $N(0, 1^2)$; recruit $N=90$, with conservative contributing $N=80$



Panel A: 20,000 simulations/cell. Panel B: 10,000 paired simulations/cell. Panels C/D: 20,000 paired simulations/scenario. Every simulated positive decision applies the stated 0.95 posterior-probability rule. Historical mean borrowing uses observed pilot Days 0, 1 and 3 only; projected Day 5 is not borrowed. Known-covariance Gaussian design approximation using pilot Day 0–3 data and a prespecified Day-5 projection. Immediate sustained effect. Dotted lines denote 80% power or 5% Type-I error.